



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)

Re-Accredited by NAAC with 'A++' Grade

Name of Student: Aryan Singh Kasera

PRN: 25070122052

Subject: Programming with Java

Sem: III AY: 2026-27

Assignment No: 6

TITLE: Write a Java program to demonstrate the use of an inner class, an anonymous class.

Problem Statement

Write a Java program to demonstrate the use of an inner class, an anonymous class.

Learning Objectives

To implement inner classes and anonymous classes for improving encapsulation and event handling in Java.

Theory

An inner class is a class defined inside another class and has access to all members of the enclosing class, including private members. Inner classes help improve code organization and logical grouping of related classes. Anonymous classes are unnamed inner classes that are instantiated only once and are mainly used for implementing interfaces or abstract classes. They simplify code by eliminating the need for separate class definitions. These concepts are widely used in Java GUI programming and event handling.

1.Create a Vehicle program where an inner class displays vehicle details and anonymousclass performs an action.

Vehicle.java



Share

Run

```
1 class Vehicle {
2     String name = "Car1";
3     int speed = 165;
4
5     class Details {
6         void display() {
7             System.out.println("Vehicle Name: " + name);
8             System.out.println("Speed: " + speed + " km/h");
9         }
10    }
11
12    interface Action {
13        void perform();
14    }
15
16    public static void main(String[] args) {
17        Vehicle v = new Vehicle();
18
19        Vehicle.Details d = v.new Details();
20        d.display();
21
22        Action a = new Action() {
23            public void perform() {
24                System.out.println("Vehicle is Running");
25            }
26        };
27
28        a.perform();
29    }
30 }
```

OUTPUT

Output

Vehicle Name: Car1

Speed: 165 km/h

Vehicle is Running

=== Code Execution Successful ===

2. Develop a Food Delivery Application where an inner class handles order details and anonymous classes handle delivery status updates.

```
FoodDelivery.java    Share  Run
```

```
1 class FoodDelivery {
2     String orderId = "20";
3     String foodItem = "Loaded Fries";
4
5     class OrderDetails {
6         void display() {
7             System.out.println("Order ID: " + orderId);
8             System.out.println("Food Item: " + foodItem);
9         }
10    }
11
12    interface DeliveryStatus {
13        void status();
14    }
15
16    public static void main(String[] args) {
17        FoodDelivery f = new FoodDelivery();
18
19        FoodDelivery.OrderDetails o = f.new OrderDetails();
20        o.display();
21
22        DeliveryStatus d = new DeliveryStatus() {
23            public void status() {
24                System.out.println("Order Delivered Successfully");
25            }
26        };
27
28        d.status();
29    }
30 }
```

OUTPUT

Output

```
Order ID: 20  
Food Item: Loaded Fries  
Order Delivered Successfully  
  
=== Code Execution Successful ===
```

Conclusion:

This experiment demonstrated the use of **inner classes** and **anonymous classes** in Java. Inner classes improve encapsulation by allowing related classes to be grouped together and providing access to the members of the enclosing class. Anonymous classes simplify code by enabling one-time implementation of interfaces or abstract classes without creating separate class definitions. Overall, these concepts enhance code organization, readability, and flexibility, and are widely used in Java applications, especially for GUI programming and event handling.